
Understanding graph representation learning in reinforcement learning based logic synthesis

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Abstract

1 Computer chips ought to be as small as possible for cost efficiency. The problem
2 of minimizing chip size while preserving the chip’s functionality has been studied
3 through the prism of and-inverter graphs (AIGs), since any Boolean function can
4 be represented by an AIG. Because the complexity of AIG size minimization is
5 unknown—it is an instance of the Minimum Circuit Size Problem, which lies
6 in NP but is not known to be NP-hard—no procedure returns optimal AIGs at
7 the sizes of interest, and there is therefore no supervision signal in the form of
8 optimal AIGs to imitate. Reinforcement learning, and more generally planning,
9 have been used to optimize AIGs in the absence of such a signal. Contemporarily,
10 graph representation learning has been applied to train machine learning models
11 to perform tasks on graph data. More recently, those two technologies have been
12 combined and reported to outperform state-of-the-art logic synthesis heuristics, but
13 the relative importance of graph representation learning in those methods remains
14 unknown. Is graph representation learning useful when optimizing and-inverter
15 graphs with reinforcement learning?

16 1 Introduction

17 A circuit that computes a given Boolean function with fewer logic gates occupies less area and is
18 cheaper to manufacture [30]. Given a Boolean function, logic synthesis methods [7, 31, 23] are used
19 to find a small circuit that computes it. Because every Boolean function can be represented with
20 AND and NOT gates only, logic synthesis methods represent circuits with and-inverter graphs (AIGs)
21 which nodes are AND gates and edges can invert incoming bits.

22 However, finding the smallest AIG encoding a given Boolean function is an open problem. It is an
23 instance of the Minimum Circuit Size Problem, which lies in NP but is not known to be NP-hard [15],
24 and which is related to Graph Isomorphism [2]. In practice, AIGs are reduced by applying sequences
25 of heuristics that rewrite the graph while preserving functional equivalence. Many such heuristics are
26 implemented in the ABC [5] software. Choosing the best heuristics sequence is a search problem.

27 In the absence of exact solvers, machine learning, such as supervised learning, has been applied to
28 AIG minimization [38]. However it suffered from two major weaknesses: first, the supervision signal
29 to train from could be a suboptimal AIG since no exact solver exists, and second, graphs generative
30 models are still quite limited [38]. Successful machine learning approaches are unsupervised ones
31 that only require a scalar signal, often the size of a given AIG which is cheap to obtain: Bayesian
32 optimization [11], multi-armed bandits [29], reinforcement learning (RL) [14, 48] and Monte Carlo
33 tree search [32]. Contemporarily, graph representation learning has been applied to train models
34 on graph data, and circuits are graphs: task-agnostic encoders built specifically for AIGs are now
35 available [18, 21, 46, 47]. The two lines of work have since been combined. Several of the methods
36 above encode the current AIG with a graph neural network rather than with a fixed set of summary

statistics [48, 32, 3], and achieve reductions beyond the expert ABC heuristics. The motivation is the same as the one behind the very first deep RL algorithm, which was designed specifically for Backgammon [40]: the state space of AIG minimization is huge and cannot be enumerated, so the value of a state has to be generalized from other states. A learned representation of the circuit should carry structural information that statistics discard, and that is what should allow a deep reinforcement learning algorithm [36] to generalize to unseen circuits: “I have seen a similar graph before, I should probably edit the AIG in a similar way to get good results”.

What those RL methods do not report is how much of their performance is attributable to the learned representation design. We are not aware of a study that isolates the contribution of the AIG encoder to RL training. Our work reports such an ablation, in the fixed-horizon setting that Zhu et al. [48] introduced and that later work reuses. We keep that pipeline fixed—the Markov decision process, the policy and value heads, the training algorithm and the hyper-parameter grid—and vary only how the AIG is presented to the policy. We show the following results.

Contributions: First, we show that adding a graph convolutional network to the policy does not improve AIG reduction beyond a multi-layer perceptron that reads graph statistics (number of nodes, of levels, . . .). Second, we derive empirical performance bounds—in terms of AIG value prediction—for message passing networks bounded by the 1-dimensional Weisfeiler-Leman refinement as well as for AIGs specific models. Next, we present existing AIG minimization methods.

2 Related work

2.1 Combinational optimization

Combinational synthesis is a class of fast heuristics that reduce an AIG by rewriting it locally, and that work well in practice [23, 24, 7]. ABC [5] is the reference implementation and is widely used both in industry and academia. Throughout this paper we call *primitives* its standard local transformations, `balance`, `rewrite`, `resub` and `refactor`, and *heuristics* the named sequences of primitives it ships, such as `resyn2`. Since no primitive is globally optimal, reduction is done by composing them into heuristics. Hand-designed heuristics are the reference points against which learned methods are measured: `resyn` [31] applies `balance`, `rewrite`, `balance`, `rewrite`, `balance`. Searching over sequences of primitives has been automated without learned circuit representations, with Bayesian optimization [11] and with multi-armed bandits [29].

2.2 Graph representation learning for AIGs

A separate line of work learns representations of circuits that are meant to transfer across circuits, rather than a policy for one circuit. DeepGate4 [47] is a foundation model that learns node embeddings of AIGs, supposedly useful for various downstream tasks on AIGs. Three encoders from this line—DeepGate [18], PolarGate [21] and FuncGNN [46]—are among the models we evaluate in section 7. GraphBench [38] is, to our knowledge, the first AIG reduction benchmark built for methods that must perform on unseen AIGs. It provides a training set of Boolean functions of varying input and output dimension and a disjoint test set, and it reports node reduction relative to the expert ABC heuristic: `resyn2`, `resyn2`, `dc2`, `resyn2rs`, `resyn2`, applied in that order. Its main result is negative: given the AIG obtained by the naive construction from a truth table, specialized DAG generative models [6, 19] do not preserve functional equivalence while reducing size.

2.3 Reinforcement learning for AIGs

The works discussed in this section use similar AIG minimization formulations: the state observed by the trained policy is some representation of the AIG to minimize and the actions are ABC primitives (sec. 2.1).

Hosny et al. [14] formulate synthesis as an MDP whose states are vectors of AIG statistics—numbers of primary inputs and outputs, nodes, edges, levels and latches, and the proportions of AND and NOT nodes—whose actions are ABC primitives, and whose reward combines node count and level count. Zhu et al. [48] keep ABC primitives as actions but encode the whole AIG with a graph convolutional network, alongside a small vector of statistics and recent actions, and reward the relative reduction

in node count. Episodes there have a fixed horizon and no early stopping, so the induced problem is a search over flows of fixed length. This is the formulation we adopt, horizon included, and we state it in full in section 4. Both Hosny et al. [14] and Zhu et al. [48] train with policy gradient methods [39, 44, 27, 36]. We are not aware of a value-based [43, 26] treatment, even though AIG editing is deterministic. These methods report minimization performances beyond the ABC heuristics, but they are not designed to generalize: a policy is trained to reduce one AIG, so the training cost is paid again for every new circuit. Search-based methods lift that restriction. AlphaSyn [32] and AlphaChip [22] plan online and therefore apply to any circuits: every AIG edit choice requires a lookahead, and every lookahead step requires network forward passes. However we argue that the cost of retraining a model is cheap compared to the cost of producing unnecessarily big chips at scale. ABC-RL [3] sits in between: a policy pre-trained on other circuits guides the tree search, and its influence is scaled down when the circuit at hand is far from anything in the training set. Two further formulations widen the setting rather than the state: ESE [34] adds technology mapping operations to the action space and trains with PPO [36], and CB-EVO [20] drops the sequential state altogether and treats the choice of primitive as a contextual bandit whose context is a set of circuit features.

2.4 Graph reinforcement learning for other combinatorial optimization

AIG reduction is an instance of graph structure optimization in the taxonomy of Darvari et al. [8], who survey reinforcement learning applied to combinatorial problems on graphs and report that a graph neural network encoder is the common choice of state representation. Within chip design the same combination is used for transistor sizing [42] and for macro placement [22]; in both cases the encoder is motivated as the component that enables transfer to new circuits. Next, we formulate AIG minimization as a Markov decision problem.

3 Background material

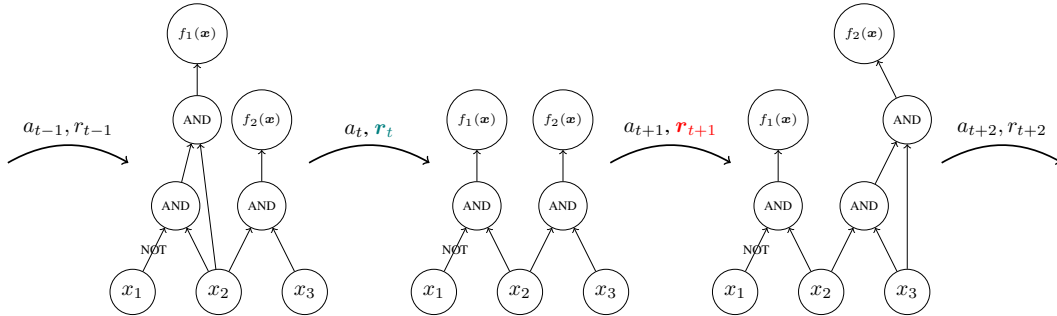


Figure 1: A trajectory in a generic MDP for AIG reduction, on AIGs of the Boolean function $f(x_1, x_2, x_3) = (\bar{x}_1 \wedge x_2, x_2 \wedge x_3)$. Every method surveyed in section 2 acts at this granularity: each action is one ABC primitive applied to the whole graph, and the primitives of section 2.1 preserve functional equivalence by construction, so every state reachable from $\mathcal{G}_0$ still computes f .

3.1 And-inverter graphs minimization

Given an arbitrary Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$, the objective is to find an and-inverter graph that encodes f with as few nodes as possible.

Section 3.2.1 from Stoll et al. [38] states that an and-inverter graph $\mathcal{G} := (V, E)$ is a directed acyclic graph of in-degree (fanin) at most two. The node set is partitioned into input nodes of in-degree 0, $V^{in} := \{v_1^{in}, \dots, v_n^{in}\}$, AND nodes of in-degree two, $V^{AND} := \{v_1^{AND}, \dots, v_k^{AND}\}$, and output nodes of in-degree one, $V^{out} := \{v_1^{out}, \dots, v_m^{out}\}$; AIGs are therefore heterogeneous graphs [37]. The edge set is $E \subseteq V \times V \times \{0, 1\}$, with 0 indicating direct transmission and 1 indicating input negation (NOT). For an input $x \in \{0, 1\}^n$, values are assigned to input nodes, propagated through gate nodes (applying inversion where specified), and collected at output nodes to yield $\mathcal{G}(x) \in \{0, 1\}^m$. Because the combination of AND and NOT is functionally complete, any Boolean function can be represented using only these two operations.

121 We write $\mathcal{G}_f := \{\mathcal{G} : \mathcal{G}(\mathbf{x}) = f(\mathbf{x}) \forall \mathbf{x} \in \{0, 1\}^n\}$ for the set of AIGs functionally equivalent to f .
 122 AIGs are non-canonical representations of Boolean functions [25]—two different AIGs can represent
 123 the same function—so $|\mathcal{G}_f| > 1$ in general: e.g. the graphs of figure 1 are functionally equivalent.
 124 Given a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$, we want to find $\mathcal{G}^* \subseteq \arg \min_{\mathcal{G} \in \mathcal{G}_f} |\mathcal{G}|$, the AIGs
 125 functionally equivalent to f with as few nodes as possible. The hardness of this problem is not known:
 126 it is an instance of the Minimum Circuit Size Problem [15], a well-studied problem in NP that is not
 127 known to be NP-hard¹.

128 3.2 Markov decision process

129 AIG minimization can be formulated as a sequential decision making task in which the environment is
 130 the current AIG, the actions are edits of that AIG, and the rewards are shaped to favor edits that remove
 131 nodes. Formally, this is the problem of finding an optimal policy for a Markov decision process
 132 (MDP) [4, 33]. An MDP is a tuple $\mathcal{M} = \langle S, A, R, T, T_0 \rangle$. S is a finite set of states representing all
 133 possible configurations of the environment (e.g. AIGs, or AIG statistics). A is a finite set of actions
 134 available to the agent (e.g. AIG edits). $R : S \times A \rightarrow \mathbb{R}$ is a deterministic reward function that assigns
 135 a real-valued reward to each state-action pair (e.g. number of deleted nodes). $T : S \times A \rightarrow \Delta(S)$ is
 136 the transition function that maps state-action pairs to probability distributions over next states $\Delta(S)$
 137 (e.g. how graph edits change the AIG; here the transitions are deterministic). $T_0 \in \Delta(S)$ is the initial
 138 distribution over states (e.g. a set of AIGs to reduce).

139 Given an MDP $\mathcal{M} \equiv \langle S, A, R, T, T_0 \rangle$, the goal of reinforcement learning is to find a
 140 policy $\pi : S \rightarrow A$ that maximizes the expected discounted sum of rewards $J(\pi) =$
 141 $\mathbb{E} [\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(s_t)) \mid s_0 \sim T_0, s_{t+1} \sim T(s_t, \pi(s_t))]$ where $0 < \gamma \leq 1$ is the discount factor that
 142 controls the trade-off between immediate and future rewards.

143 We write $V^*(s) = \mathbb{E} [\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(s_t)) \mid s_0 \sim s, s_{t+1} \sim T(s_t, \pi(s_t))]$ for the value of a state
 144 under an optimal policy, i.e. the discounted sum of rewards collected from s by acting optimally.
 145 V^* is the quantity a critic is trained to approximate in actor-critic methods, and the quantity a state
 146 representation must be able to distinguish states by; section 7 uses it to measure what a representation
 147 can express, independently of any training procedure.

148 3.3 Message passing neural networks

149 A message passing neural network (MPNN) [10] attaches a vector $h_v^{(0)}$ to each node v , initialized from
 150 that node’s features, and refines it over k rounds: $h_v^{(t+1)} = \phi^{(t)}(h_v^{(t)}, \bigoplus_{u \in \mathcal{N}(v)} \psi^{(t)}(h_v^{(t)}, h_u^{(t)}, e_{uv}))$,
 151 where $\mathcal{N}(v)$ are the neighbours of v , e_{uv} is an optional label on the edge between u and v , $\bigoplus$ is a
 152 permutation-invariant aggregator, and $\phi^{(t)}, \psi^{(t)}$ are learned. A graph-level vector is then obtained
 153 by pooling $\{h_v^{(k)}\}_{v \in V}$ with a second permutation-invariant readout. Architectures in this family
 154 differ only in the choice of ψ , $\bigoplus$ and ϕ : GCN [17], GIN [45], GAT [41] and GraphSAGE [13] are
 155 instances, and so are the AIG-specific encoders we evaluate in section 8. They also share a limitation.
 156 After k rounds, two nodes that k iterations of the 1-dimensional Weisfeiler-Leman refinement assign
 157 the same color carry the same embedding [45, 28, 1], so no network in the family can tell apart two
 158 AIGs that 1-WL cannot. Section 7.2 turns this into a lower bound on the error of every such encoder.

159 Next, we present our methodology for studying the effect of MPNNs when using RL to train policies
 160 for AIG minimization formulated as an MDP.

161 4 Methodology

162 We use the same methodology as Zhu et al. [48] who first combined graph convolutional networks and
 163 (deep) reinforcement learning for AIG minimization. We use a similar MDP to their’s: the actions are
 164 the five ABC primitives {balance, rewrite, rewrite -z, refactor, refactor -z}, transitions
 165 are deterministic and every primitive preserves functional equivalence, so every reachable state stays
 166 in $\mathcal{G}_f$, and an episode applies exactly $H = 20$ of them to the AIG $\mathcal{G}_0$ to reduce, which makes the
 167 induced optimization a search over flows of length 20 from a five-letter alphabet. The reward differs

¹<https://mcsp.work/index.html>

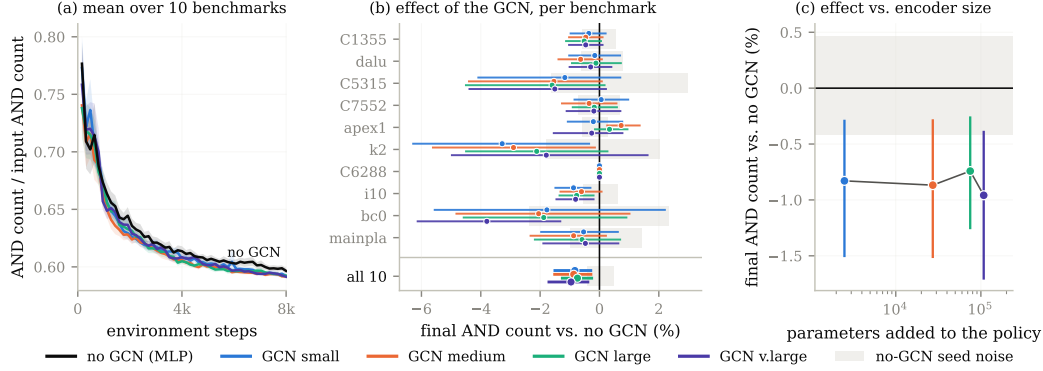


Figure 2: PPO on the ten MLCAD benchmarks with five observation encoders, each at its own best hyper-parameters. (a) AND count averaged over the ten benchmarks; (b) change in final AND count against the no-GCN policy, per benchmark and pooled over the ten (*all 10*), the grey band being the no-GCN policy’s own spread over its five seeds; (c) the pooled change against the parameters the encoder adds to the policy. All intervals are 95% bootstrap confidence intervals over five seeds.

slightly: writing n_t for the number of AND nodes after t actions, $R(s_t, a_t) = 1 - n_{t+1}/n_0$ is issued at every step, so with $\gamma = 1$ we optimize node count alone—the objective of section 3.1—and not the multi-objective reward of Hosny et al. [14]. Unlike Zhu et al. [48], we use PPO [36] to train policies instead of REINFORCE [44] since the former is the better more modern learner.

The policies we train always read a features vector $x_G \in \mathbb{R}^{|A|+5}$ holding the histogram of the actions taken so far, the AND and level counts of the current and previous AIG relative to $\mathcal{G}_0$, and the normalized time step: this is the same as in Hosny et al. [14], Zhu et al. [48]. This features vector carries no AIG structural information, since two AIGs with the same size, depth and history are indistinguishable to it. The key novelty of Zhu et al. [48] is to feed the policy a graph convolutional network [17] reading the AIG and pooling it into a graph-level vector concatenated with the features vector x_G . This GCN addition has since then been built upon (cf section 2.3) but never ablated against e.g. Hosny et al. [14]. Ablating GCN, which is the focus of the following sections, is important because while they might increase performances, they definitely add memory and time overheads.

5 Graphs reinforcement learning: reproducing Zhu et al. [48]

5.1 Setup

We minimize the same AIGs as Zhu et al. [48]: apex1, bc0, C1355, C5315, C6288, C7552, dalu, i10, k2 and mainpla. They range from 504 to 6359 AND gates and encode diverse Boolean functions. For each AIG we run PPO with five policies: no GCN, and a small, medium, large and very large GCN. Every (AIG, policy) pair is tuned over the same grid — three learning rates by three entropy coefficients, five seeds each — and we report the cell with the lowest mean final AND count over its five seeds, discarding cells in which a seed did not complete. Each policy therefore receives the same tuning budget and is shown at its own optimum. We periodically save and evaluate the greedy PPO policy: we use it to minimize an AIG and report the AND count after applying $H = 20$ ABC primitives (cf. section 4). All hyperparameters and implementation details are given in appendix A.1.

6 Results

Figure 2(a) shows that a tuned PPO learns to minimize an AIG independently of the policy architecture, reaching a median performance 6% below resyn2 [31]. Figure 8 in appendix E breaks the same runs down by benchmark and the observation holds. Pooled over the ten benchmarks, last row of figure 2(b), the GCN policies do end between 0.74% and 0.96% below the multi-layer perceptron (MLP) policy with confidence intervals excluding zero, and the best GCN policies produces the

smaller circuits on eight to nine benchmarks out of ten (see appendix table 2). The effect is nonetheless marginal: it is barely larger than what re-running the baseline with different seeds produces (grey band), only 6 of the 40 encoder-benchmark comparisons have a confidence interval entirely below zero, and in absolute terms the best GCN saves a median of 11 AND gates per benchmark (see appendix table 2), from none at all on C6288 to 57 gates on bc0. Most importantly, figure 2(c) shows the gain does not increase with encoder size: 2 496 added parameters improve performances as much as 107 968. We repeated the experiment on the 100 AIGs (106 to 46 977 AND gates) of the 2026 IWLS programming contest². The picture there is the same in sign and magnitude, 0.3% to 0.7% below the MLP (appendix B).

Next, we ask why the grains from GCN policies are so marginal despite being a representation specialized for graphs compared to MLPs reading graphs statistics.

7 Why is graphs representation learning improving so little over MLPs?

The previous section compares encoders inside one algorithm, so a null result there could be a property of that algorithm rather than of the representations. We therefore ask a question that does not involve a training algorithm at all: how well can the optimal value V^* of a state (cf section 3.2) be predicted from what a given class of models is able to see? This empirical analyses of graphs representation, in particular of MPNNs (cf. section 3.3), has been studied for graph-level regression tasks other than value prediction in Akbiyik et al. [1].

7.1 Building a dataset

Computing V^* exactly is out of reach: the tree of flows over the five primitives of section 4 has 5^H leaves, and every edge of it costs one ABC call on the full circuit. We therefore estimate it by sampling both the states and the continuations from them. A state is the AIG obtained by applying to the benchmark $\mathcal{G}_0$ a uniformly random flow of length $L \sim \mathcal{U}\{2, \dots, 10\}$; we draw 1,000 such states per benchmark and store each one both as an .aig file and as the statistics $x_{\mathcal{G}}$ of section 4, so that the graph models and the MLP of sections 7.2 and 8 are evaluated on exactly the same states and the same targets. From each state s we then run $M = 100$ independent rollouts of 8 further uniformly random primitives and keep the best AND count reached, $\hat{V}(s) = \min_{j \leq M} n^{(j)}(s)$, an upper bound on the minimum over the 5^8 flows of length eight that approaches it from above as M grows. Two differences to V^* should be kept in mind. $\hat{V}$ counts AND nodes rather than returns, so it is ordered the other way round from the value of section 3.2—smaller is better—which leaves the comparison between model classes unchanged (appendix A.2). And it truncates the horizon at eight steps instead of the $H = 20$ of the reinforcement learning experiments, so our targets approximate the value of an eight-step-to-go problem, and the bounds of section 7.2 are lower bounds on the error of predicting *that* quantity. The procedure is repeated with ten seeds per benchmark on the same ten MLCAD circuits as in section 5, and the figures of this section report over those ten repetitions. The sampler is degenerate on two of them, reaching only two distinct values of $\hat{V}$ on C6288 and six on C1355: we exclude C6288 and report over the remaining nine benchmarks, flagging C1355 wherever it appears.

7.2 Theoretical lower bounds on the prediction error

A model that cannot distinguish two states must assign them the same prediction, so a class of models partitions the state space and its best achievable squared error is that of the predictor returning, for each part, the mean target over that part. No model in the class can beat that error, regardless of its size, its training procedure or its optimization. This is the protocol of Graphtester [1], which measures the theoretical boundary of MPNNs (cf. section 3.3) on a given dataset rather than the performance of one trained instance; we reimplement it for the target of section 7.1 and for four representations, ordered by what they can see: the number of nodes $|V_{\mathcal{G}}|$ alone, which is what a 0-layer message passing network reduces to; the pair $(|V_{\mathcal{G}}|, d_{\mathcal{G}})$, with $d_{\mathcal{G}}$ the logic depth reported by ABC; the engineered statistics $x_{\mathcal{G}}$ of section 4 in full, the action histogram included, which is exactly what the MLP policy from section 5 reads; and the 1-WL coloring of the AIG, which upper bounds the separating power of every architecture of section 3.3. Comparing the last two is the question of this section: what a learned encoder could express, against what the statistics already

²<https://github.com/alanminko/iwls2026-ls-contest/>

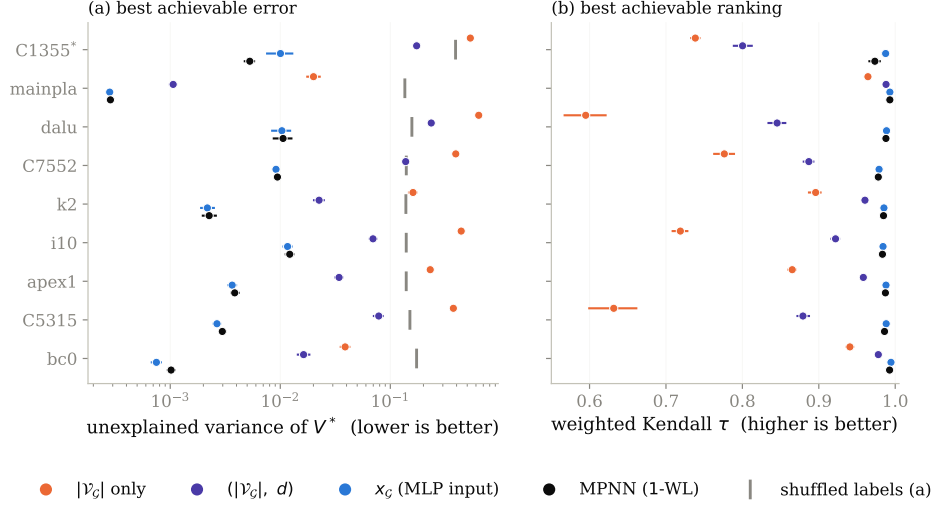


Figure 3: Best error (a) and best ranking (b) achievable by each model class on $\widehat{V}$, over nine benchmarks and ten dataset seeds. Each class is replaced by the optimal predictor for the information it can distinguish, so the value shown cannot be beaten by any model in that class. *C1355 has only six distinct targets.

express. They are incomparable rather than nested—only the current size and depth entries of x_G are functions of the current AIG, the previous step’s size and depth, the normalized time step and the action histogram being episode history that no encoder of the current graph can recover—so the x_G bound may fall below the 1-WL bound without contradicting it, and a policy that reads both, as Algorithm 1 does, is bounded by neither alone. Besides the squared error, which we normalize by the variance of the target, we report how well each class can *order* states, with Spearman’s ρ , Kendall’s τ and the weighted τ , since a critic that ranks successors correctly is useful even when its absolute errors are large. The bounds are computed in sample, so a partition fine enough to isolate every state would drive the error to zero regardless of whether the representation carries any signal about $\widehat{V}$; we control for this by recomputing every bound with the targets randomly permuted across states, which leaves the partition intact and destroys the association. Every bound we report sits 11 to 474 times below its shuffled-label counterpart, depending on the benchmark, and the two partitions we compare are of nearly equal granularity, at 865 parts for x_G and 860 for 1-WL out of 1,000 states on average, so neither class is merely memorizing the targets nor given more degrees of freedom than the other. Appendix C details how the partitions are computed.

7.3 Results

A k -layer MPNN separates graphs at most as finely as k iterations of 1-WL, so grouping states by their 1-WL color and predicting each group’s mean $\widehat{V}$ gives an error no MPNN can beat; the same construction bounds a 0-layer network, a (size, depth) model, and a model seeing the MLP’s features x_G . From figure 3, we observe that message passing is worth a great deal over global descriptors: the bound falls from 0.02–0.64 of the target variance for node count alone, to 1.1×10^{-3} –0.24 once depth is added, and to 3×10^{-4} – 1.2×10^{-2} for 1-WL. It is worth nothing over x_G . On eight of the nine benchmarks the x_G bound is the *lower* of the two, at 0.73 to 0.99 times the MPNN bound; only C1355 goes the other way, where message passing is better by $1.93\times$. The ordering is stable within a dataset: paired by seed, x_G is lower on all ten seeds of six benchmarks and on seven and eight of ten for dalu and C7552. For ranking the two classes are indistinguishable — weighted Kendall $\tau \geq 0.97$ for both on every benchmark, with x_G ahead by at most 0.014 — while size and depth alone reach only 0.60–0.99. This is possible because x_G is not a coarsening of the graph: it carries the episode history described above, which no encoder of the current AIG can reconstruct. Depth is irrelevant — the bound moves by at most 1.1×10^{-3} between $k = 1$ and $k = 50$, and edge direction and inverter bits change it by less than 10^{-6} (Appendix D). C6288 is excluded: its target takes two distinct values.

Limitations of the bounds. Four restrictions should be read with the numbers above. First, they hold over the state distribution of section 7.1—AIGs reached by uniformly random flows of length two to ten—and not over the states a trained policy visits, which are fewer and more correlated; a representation could be uninformative on the former and useful on the latter. Second, each dataset descends from a single benchmark, so what is bounded is the ability to tell apart states *of one circuit*, whereas the argument for learned encoders in section 4 is transfer *across* circuits; our experiment does not test that claim, and a cross-circuit dataset could behave differently. Third, $\hat{V}$ is a Monte-Carlo estimate, so part of its variance is sampling noise that no representation can predict. That noise raises every bound by roughly the same amount and therefore compresses the ratios between classes towards one, which works in favor of the negative conclusion we draw; the 1-WL and x_G bounds being close is evidence that message passing adds little, but not proof that the noise floor is not what they are both close to. Fourth, the 1-WL argument covers message passing networks whose input is the AIG with node-type features, which is the setting of section 4 and of Zhu et al. [48]. It does not cover higher-order architectures, models given node identifiers or positional encodings, nor encoders whose node features are not a function of the graph alone—the simulation-derived features used by some AIG-specific models are an example, and for those the bound of figure 3 is not binding.

In the next section, we train models to predict the target values of section 7.1.

8 Empirical performances of AIG-specific models on value prediction

The bounds of section 7.2 say what a representation *could* express. This section asks what models actually trained on the same datasets achieve, which is the quantity a critic in Algorithm 1 would reach.

8.1 Setup

We train eight regressors on the datasets of section 7.1. Four are generic message passing networks—GCN [17], GIN [45], GAT [41] and GraphSAGE [13]—taken from PyTorch Geometric [9]. Three are AIG-specific: DeepGate [18], PolarGate [21] and FuncGNN [46], each run from its authors’ implementation. The eighth is the MLP of section 4, which reads x_G and no graph at all, and is the baseline the graph models have to beat. All eight share the same head so that only the encoder differs: node embeddings are mean-pooled into one vector per AIG, which a two-layer ReLU network maps to a scalar. GCN, GIN and GraphSAGE consume the six-way one-hot node type of section 4; GAT, PolarGate and FuncGNN consume the three-way type (input, AND, inverter) of the DeepGate reader, with the inverter bit exposed as a signed edge feature, so that the models that were designed to use complementation receive it. DeepGate is the one exception to end-to-end training: we load the pretrained encoder, freeze it, and train only the head on its mean-pooled functional embeddings, which is how it is meant to be used as a general-purpose circuit encoder.

Targets are $\hat{V}$ divided by the AND count of the benchmark’s initial AIG, so they lie in $(0, 1]$ and are comparable across circuits. Each dataset of 1,000 states is split 80/10/10 into training, validation and test sets by a permutation seeded by the run seed. Models are trained for 1,000 epochs with Adam [16] on the mean squared error. Each architecture is given its own grid of 36 configurations, and 12 for DeepGate, whose frozen encoder leaves only the head to tune. The grid is run at seed 0, the configuration with the lowest final validation error is selected, and that configuration alone is then rerun on the ten seeds we report; the seed selects the dataset, the split and the initialization together, so the ten repetitions are independent in all three. Appendix A.2 gives the grids. We report test error normalized by the variance of the test targets, so that 1.0 is the error of predicting the mean.

8.2 Results

All seven graph models land between 0.90 and 1.74 times the target variance — at or above the constant predictor — while the MLP reaches 0.47 at the median. Their training error is also ≈ 1 , so this is a failure to fit rather than to generalize, and it leaves them two to three orders of magnitude above the bound of figure 3 that their own architecture class admits. The per-dataset validation curves of figure 9 in appendix E show the graph models settling at the level of the constant predictor early in training and staying there.

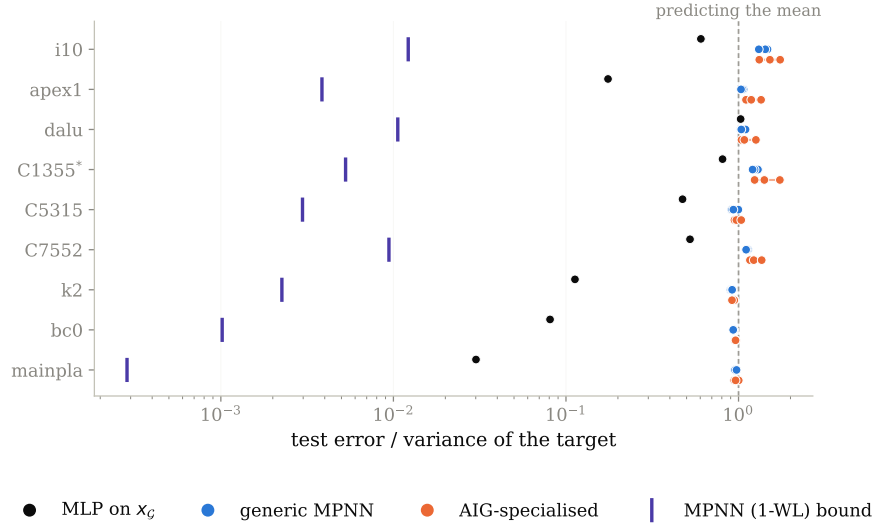


Figure 4: Test error of eight trained models, normalized by the variance of the target so that 1.0 is the error of predicting the mean. Violet ticks: the 1-WL bound of figure 3.

9 Future work

Our results are restricted to the setting used by past work: a policy gradient algorithm, five ABC primitives as actions, and the fixed horizon of section 4. Two of those choices—the algorithm and the action space—bear directly on how much a learned representation of the circuit could contribute, and relaxing them is where we would look next.

Other reinforcement learning algorithms. AIG editing is deterministic and its exploration problem is hard, a combination for which value-based algorithms [43, 26] are usually competitive, yet the existing work on AIG reduction surveyed in section 2 is entirely policy gradient-based. That distinction matters here rather than only for sample efficiency: a value-based agent acts by comparing the predicted values of successor states, so its behavior rests entirely on the quantity section 7 measures, whereas in an actor-critic method the value estimate only shapes the advantage. If the representation is the bottleneck anywhere, it should be there, and our experiments do not answer whether a graph encoder helps under such an algorithm.

The full ABC action space. Five primitives give the policy little to decide, and the states it can reach are the images of a handful of short sequences, which may be why the statistics of section 4 suffice to tell them apart. ABC exposes many more primitives, and our environment already implements that larger action space (Section 4) even though the experiments above do not use it. With more primitives, reachable states would differ from one another in more ways, and a representation that sees the circuit rather than its summary would have correspondingly more to distinguish. Rerunning both the ablation of section 5 and the bounds of section 7.2 over that action space would say whether our negative result is a property of AIG reduction itself or of the small action space it is usually given.

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A Hyper-parameters

A.1 Graph encoders and reinforcement learning

The environment. The MDP of section 4 instantiates the MDP of section 3.2 with the five ABC primitives as actions, where `-z` enables zero-cost replacements. Episodes start from $\mathcal{G}_0$ and last exactly $H = 20$ actions with no early stopping. Because every primitive preserves functional equivalence, the problem of section 3.1 is never violated, only under-solved. The reward is a level rather than a difference, so the return is maximized by an agent that keeps the AIG small throughout the episode and not merely at the end. Our environment also implements a seven-action variant adding `resub` and `resub -z`, and the full ABC action space, neither of which is used in our experiments.

The observation. Writing n_t and ℓ_t for the number of AND nodes and of levels after t actions, the vector part of the observation is

$$x_{\mathcal{G}} = \left[\underbrace{\frac{1}{q} \sum_{j=1}^q \mathbf{1}[a_{t-j} = a]}_{a \in A}, \frac{n_t}{n_0}, \frac{\ell_t}{\ell_0}, \frac{n_{t-1}}{n_0}, \frac{\ell_{t-1}}{\ell_0}, \frac{t}{H} \right].$$

The first $|A| = 5$ entries are a normalized histogram of the last q actions, with $q = 3$ like in Zhu et al. [48]; the next four are the AND and level counts of the current and of the previous AIG, each divided by its value in $\mathcal{G}_0$ so that the vector is scale-free across benchmarks of very different sizes; the last is the normalized time step, which makes the finite-horizon problem Markov in the observation. It is computed in a single pass over the graph and adds no parameters of its own. The graph part of the observation is the AIG itself, as in Zhu et al. [48]: nodes carry a one-hot encoding of their type among the six types `abc_py` exposes, a directed edge runs from each fanin to the node it feeds, and edges carry no label.

The encoders. Table 1 lists the four graph encoders of section 4: three at depth three with widths from 12 to 128, and one at depth five, so that depth and width are not confounded. Each runs the message passing rounds of section 3.3 over the AIG and pools the node embeddings into a graph-level vector concatenated with $x_{\mathcal{G}}$ before the policy and value heads. The baseline uses no encoder at all and reads $x_{\mathcal{G}}$ alone through a multi-layer perceptron.

Table 1: The four graph encoders. Depth is the number of message passing rounds, width the hidden dimension, and output the dimension of the pooled vector concatenated with $x_{\mathcal{G}}$.

encoder	depth	width	output
<code>small</code>	3	12	4
<code>medium</code>	3	64	32
<code>large</code>	3	128	64
<code>v. large</code>	5	128	64

Training. Policies are trained with the PPO implementation of `stable-baselines3` [35] with $\gamma = 1$, the horizon being finite and fixed, and the encoder is trained end to end with the policy and value heads rather than pre-trained or frozen. Eight environments run in parallel and each contributes 20 steps per update, so an update is computed from 160 transitions and a single episode spans five updates; 8,000 environment steps therefore amount to $N = 50$ updates. After every update we evaluate the greedy policy for one episode, which is the quantity the learning curves report. The policy and value heads are each two hidden layers of 256 units and share the encoder of line 7 of Algorithm 1. Algorithm 1 states the resulting procedure, which we call `GRAPHPPPO`: it is ordinary PPO with the encoder made explicit, the encoding of the current AIG being the only place where the five configurations we compare differ.

The grid of section 5 crosses three learning rates (10^{-4} , 10^{-5} , 10^{-6}), three entropy coefficients (0, 0.01, 0.1) and five seeds, which with five encoders and ten benchmarks gives the 2,250 runs reported there. Twenty-six of those runs (eighteen `small`, eight `v. large`) stopped before the fiftieth update; we drop them rather than pad them, and requiring all five encoders to have completed a given (benchmark, learning rate, entropy, seed) cell leaves 424 of the 450 cells.

Algorithm 1 GRAPHPPPO. The encoder f_θ is the only component that varies across the runs we compare; everything else is held fixed.

Require: AIG $\mathcal{G}_0$ to reduce with n_0 AND nodes, encoder f_θ , policy head π_θ , value head V_θ , action set A of five ABC primitives, fixed horizon $H = 20$, iterations N , epochs K , clipping ϵ , entropy weight β , value weight c , step size η

```

1:  $n^* \leftarrow n_0$ 
2: for  $i = 1, \dots, N$  do
3:    $\mathcal{D} \leftarrow \emptyset$ 
4:   for each parallel environment do
5:      $\mathcal{G} \leftarrow \mathcal{G}_0$ 
6:     for  $t = 0, \dots, H - 1$  do
7:        $s_t \leftarrow (f_\theta(\mathcal{G}), x_{\mathcal{G}})$   $\triangleright f_\theta$  is a GCN, or is absent for the baseline
8:        $a_t \sim \pi_\theta(\cdot | s_t), a_t \in A$ 
9:        $\mathcal{G} \leftarrow \text{ABC}(\mathcal{G}, a_t)$   $\triangleright$  deterministic, preserves functional equivalence
10:       $r_t \leftarrow 1 - |V^{AND}(\mathcal{G})| / n_0$   $\triangleright$  relative size, issued at every step
11:       $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, a_t, r_t, \log \pi_\theta(a_t | s_t), V_\theta(s_t))\}$ 
12:       $n^* \leftarrow \min(n^*, |V^{AND}(\mathcal{G})|)$ 
13:    end for
14:  end for
15:  compute returns  $\hat{R}_t$  and advantages  $\hat{A}_t$  from  $\mathcal{D}$ 
16:   $\theta_{\text{old}} \leftarrow \theta$ 
17:  for  $K$  epochs over minibatches of  $\mathcal{D}$  do
18:     $\rho_t(\theta) \leftarrow \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ 
19:     $L(\theta) \leftarrow \mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t) - c(V_\theta(s_t) - \hat{R}_t)^2 + \beta \mathcal{H}[\pi_\theta(\cdot | s_t)]]$ 
20:     $\theta \leftarrow \theta + \eta \nabla_\theta L(\theta)$   $\triangleright$  gradients flow through  $f_\theta$ 
21:  end for
22: end for
23: return  $n^*$ , the smallest AND count visited

```

benchmark	input	no GCN (MLP)	GCN small	GCN medium	GCN large	GCN v. large	resyn2
C1355	504	389 $\pm$ 3	387 $\pm$ 1	387 $\pm$ 0	387 $\pm$ 1	387 $\pm$ 0	390
dalu	1740	1008 $\pm$ 9	1006 $\pm$ 6	1001 $\pm$ 3	1006 $\pm$ 6	1005 $\pm$ 2	1052
C5315	1773	1303 $\pm$ 43	1287 $\pm$ 8	1283 $\pm$ 2	1282 $\pm$ 6	1283 $\pm$ 5	1309
C7552	2074	1371 $\pm$ 12	1372 $\pm$ 10	1367 $\pm$ 11	1369 $\pm$ 5	1369 $\pm$ 10	1455
apex1	2124	1066 $\pm$ 7	1064 $\pm$ 11	1074 $\pm$ 4	1070 $\pm$ 4	1063 $\pm$ 14	1179
k2	2190	1049 $\pm$ 26	1015 $\pm$ 32	1019 $\pm$ 27	1027 $\pm$ 21	1030 $\pm$ 36	1194
C6288	2337	1870 $\pm$ 0	1870 $\pm$ 0	1870 $\pm$ 0	1870 $\pm$ 0	1870 $\pm$ 0	1870
i10	2675	1715 $\pm$ 13	1700 $\pm$ 3	1705 $\pm$ 9	1702 $\pm$ 3	1702 $\pm$ 6	1829
bc0	3501	1512 $\pm$ 46	1485 $\pm$ 59	1481 $\pm$ 31	1484 $\pm$ 28	1455 $\pm$ 13	1713
mainpla	6359	2318 $\pm$ 37	2306 $\pm$ 14	2298 $\pm$ 11	2305 $\pm$ 23	2307 $\pm$ 14	2476
mean AND/input		0.596	0.592	0.592	0.592	0.591	0.628

Table 2: Mean AND count after $H = 20$ ABC primitives over the five seeds of each configuration’s best hyper-parameter cell, $\pm$ one standard deviation across those seeds; best per benchmark in bold. resyn2 is deterministic. The closing row normalises by the input AND counts.

544 A.2 Value prediction

545 **The datasets.** The states of section 7.1 are drawn by sampling L primitives independently and
546 uniformly from the five-element action set and running them in order; the graph models read the
547 stored .aig file and the MLP reads the statistics $x_{\mathcal{G}}$ recorded at that point of the flow. The state is
548 reloaded between the M rollouts. Producing one dataset costs $1,000 \times (L + M \times 8) \approx 8 \times 10^5$ ABC
549 primitive calls per benchmark and per seed, which is why M is lowered to 10 on C6288, by far the
550 most expensive circuit of the suite to rewrite. Each of the ten seeds (0 to 9) draws its own 1,000 states
551 and its own rollouts, and each resulting dataset is split at random into training, validation and test
552 sets. That the target is expressed in AND nodes rather than in returns does not affect the comparison
553 between model classes: the map $n \mapsto 1 - n/n_0$ from AND count to the reward of section 4 is affine

554 and order-reversing, all the errors we report are ratios to the variance of the target and are therefore
555 free of its units, and the rank correlations we report are unchanged up to sign.

556 **Training.** The models of section 8 are trained in mini-batches of 16 unless the batch size is itself
557 searched over. Each architecture’s grid of 36 configurations crosses three learning rates (10^{-3} ,
558 5×10^{-4} , 10^{-4}) with a width and a depth, or with the number of attention heads for GAT and the
559 polarity weight for PolarGate. DeepGate is the exception, with 12 configurations: its pretrained
560 encoder is frozen, so only the two-layer head is tuned.

561 B The ex200–ex299 suite

562 We repeat the comparison of section 5 on ex200–ex299, a suite spanning 106 to 46,977 input AND
 563 gates and therefore both larger and more varied than the ten MLCAD circuits. Each encoder ran at its
 564 own baseline hyper-parameters here instead of over the matched grid. The three GCNs reduce the
 565 final AND count by 0.33–0.71% against a reseed standard deviation of 0.54% (Figure 5), and the
 566 difference is flat across two and a half orders of magnitude of circuit size (Figure 6), so the conclusion
 567 of section 5 appears to hold here as well. The MLP runs were executed on CPU and did not finish on
 568 the largest circuits, which is why 20 of the 100 benchmarks are excluded; the comparison is therefore
 569 made on the smaller eighty, and we cannot rule out that the largest circuits behave differently.

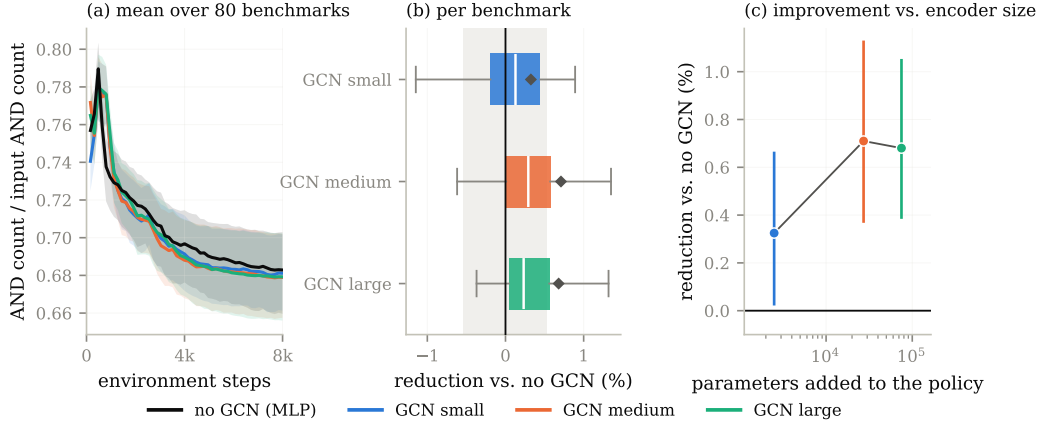


Figure 5: The comparison of figure 2 on ex200–ex299, where each encoder ran at its own baseline hyper-parameters instead of the matched grid. Restricted to the 390 of 500 (benchmark, seed) cells all four encoders completed, covering 80 benchmarks.

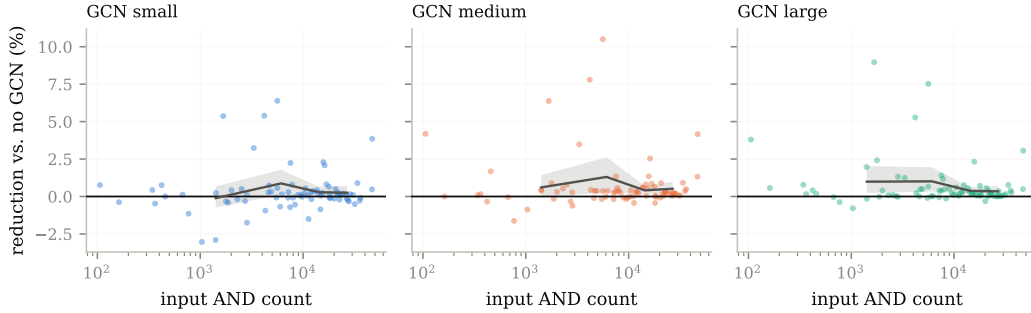


Figure 6: Reduction against the MLP as a function of input circuit size on ex200–ex299. Line: mean over size quartiles.

570 C Computing the bounds

571 For the size and the size-and-depth classes of section 7.2 the partition is by exact equality of the
 572 descriptors. For x_G , the full feature vector is snapped to a grid of spacing 10^{-8} and states are grouped
 573 by equality of the snapped vector, so the model we bound and the MLP of section 4 see the same thing.
 574 For message passing, we group states by their 1-WL graph hash, computed with `networkx`’s [12]
 575 `weisfeiler_lehman_graph_hash` at every number of iterations $k = 1, \dots, 50$, with each node
 576 labeled by its type.³ We bound four views of the AIG, which is what the four message passing curves
 577 of figure 3 report: the undirected graph $\mathcal{G}$, the directed graph $\mathcal{G}^{\rightarrow}$ in which refinement follows the
 578 fanin relation, and the two variants $\mathcal{G}^e, \mathcal{G}^{\rightarrow e}$ that additionally label each edge with its complement
 579 bit, so that inversion is visible to the refinement rather than only through node types. Errors are
 580 divided by $\max(1, \text{Var } \hat{V})$; in AND-node units the variance is orders of magnitude above one on
 581 every benchmark, so the clamp never binds and the reported quantity is the plain normalized error.
 582 Rank correlations are computed at $k = 50$ refinement iterations. The permutation control uses 25
 583 permutations per dataset, 8 on `mainpla` and `C6288`, and is applied to the rank correlations as well,
 584 where the null is the correlation between $\hat{V}$ and a random permutation of itself.

³The hash is a 16-byte `blake2b` digest. A collision would merge two groups that 1-WL separates and can only raise the reported value, so the bound would become conservative rather than optimistic; at 1,000 graphs per dataset the probability of one is negligible.

585 D The message passing bound against depth

586 Figure 7 plots the message passing bound of section 7.2 against the number of refinement iterations k ,
 587 for the four views of the AIG bounded there. The bound moves by at most 1.1×10^{-3} between $k = 1$
 588 and $k = 50$, and labeling edges with direction or with the complement bit changes it by less than
 589 10^{-6} , so neither deeper refinement nor a richer view of the graph brings a message passing network
 590 closer to $\hat{V}$.

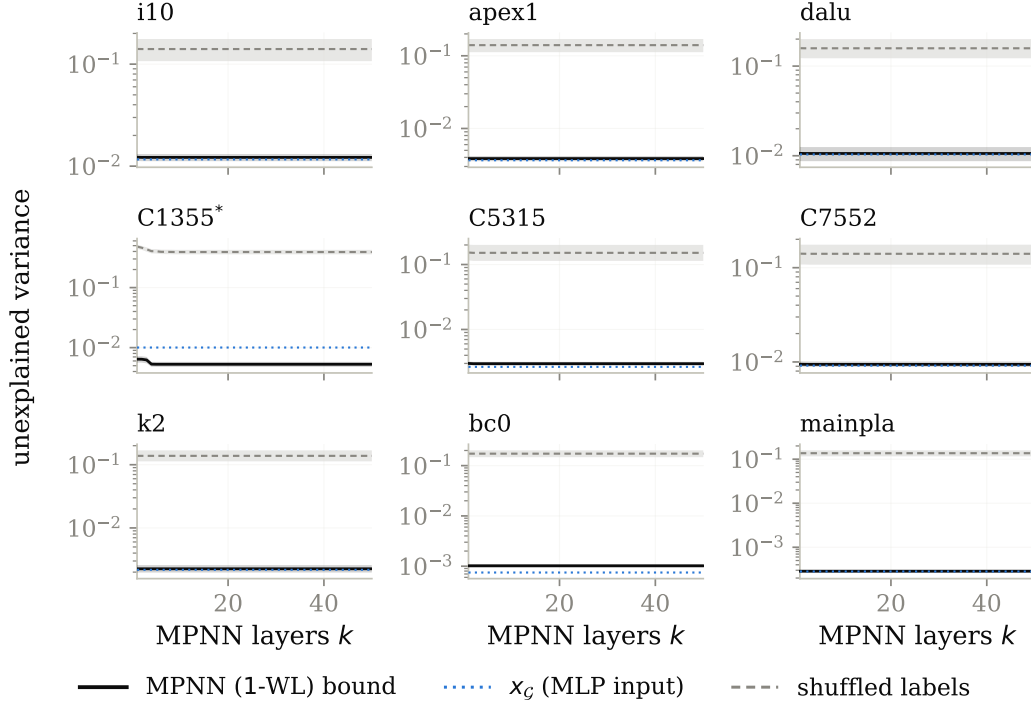


Figure 7: The MPNN bound as a function of depth k .

591 E Per-dataset curves

592 Figure 8 resolves the aggregate of figure 2(a) into one learning curve per benchmark, and figure 9
 593 shows validation error over training for the value prediction models of figure 4.

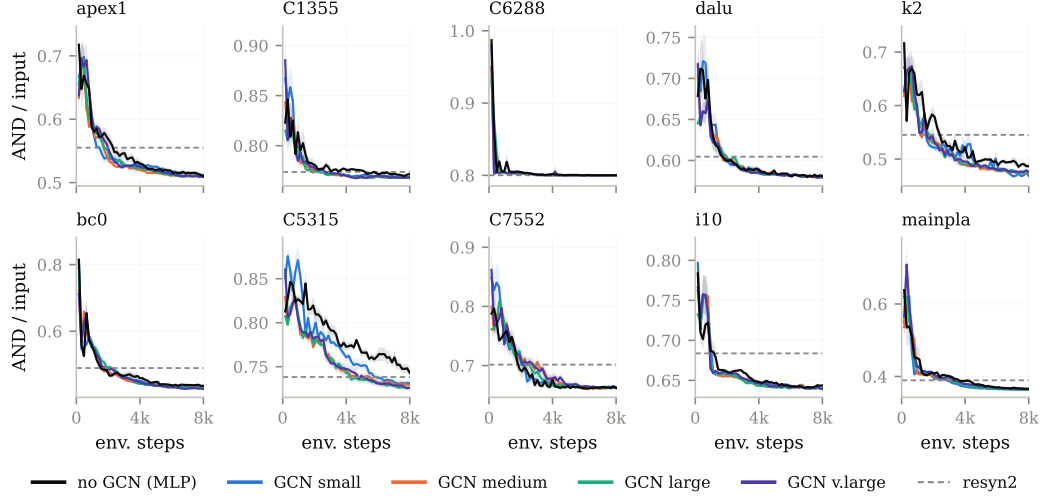


Figure 8: Per-benchmark learning curves for the runs aggregated in figure 2(a). Dashed line: resyn2.

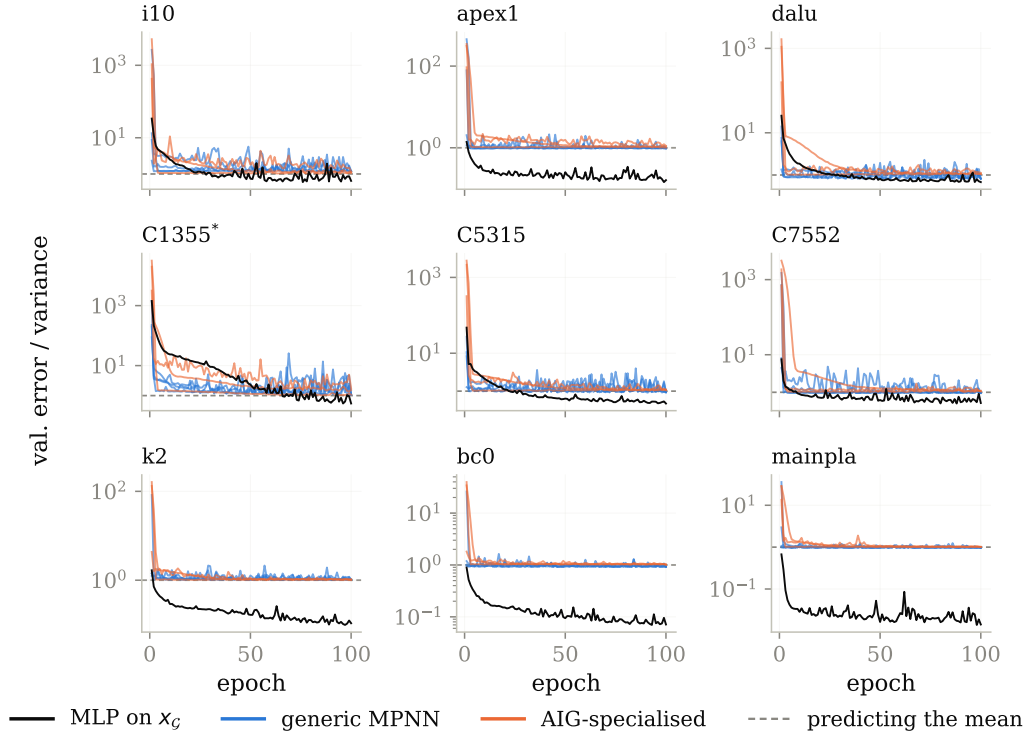


Figure 9: Validation error over training for the configurations of figure 4, one panel per benchmark.